COMP 490 Final Report - Fall 2025

Project Title: *Investigating Approaches for Implanting Backdoor Triggers into Deep Neural Networks.*

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GitHub Repository: https://github.com/amannuck/comp490_backdoor_attacks

Abstract

Deep Neural Networks (DNNs) used in time-series analysis are increasingly used in sensitive domains such as healthcare and electrical grid monitoring, but they remain vulnerable to backdoor attacks where a model behaves normally on clean data but produces attacker-controlled outputs when a hidden trigger is present. This project investigates how such backdoors can be created, implanted, and evaluated across different types of time-series models. We started by reproducing generative poison-label attacks (TSBA) on eight univariate datasets to study how dataset structure, class count, and how noise affects the success of trigger-based attacks. We then apply a stealthy trigger detector to ECG5000, showing that a small network monitoring a short region of the signal can override a clean classifier while causing no change in accuracy. We further extend this approach to a multivariate Automatic Generation Control (AGC) dataset by combining a Random Forest classifier with a trigger detector trained on short AGC windows. Across all experiments, the backdoors achieved high attack success rates with minimal drops in clean accuracy, demonstrating strong stealthiness. The AGC results reveal that an attacker can force attack sequences to be classified as normal, which poses a serious security risk for power-system monitoring. The study shows that backdoor attacks generalize across domains, from biomedical data to power-system signals, and across different architectures, including deep neural networks and classical machine-learning models, highlighting the broader vulnerability of time-series systems to hidden trigger manipulation.

1. Introduction

Deep Neural Networks (DNNs) are widely used in critical systems such as healthcare [4], finance [5], and electrical grids [6], often used to analyze time-series data. Time-series data is a data type where we capture a sequence of observations with measurable quantities indexed by timestamps. [1] However DNNS are exposed to security vulnerabilities such as backdoor attacks. Backdoor attacks are one of the serious threats to DNNs whereby a model behaves normally on clean data but misclassifies inputs that contain a hidden trigger, implanted by the attacker at training time. Since the trigger rarely occurs during normal operation, the malicious behavior can remain undetected. This project aims to study and implement different methods of implanting backdoor triggers in deep neural networks, focusing on their effectiveness, stealth, and adaptability across domains such as time-series data.

1.1 Background and Motivation

Prior studies, such as Jiang *et al.* [1] and Tang *et al.* [2], have explored simple and generative approaches to backdoor injection. These studies reveal that time-series models can be manipulated through subtle waveform changes which could pose risks for real-world systems that monitor physical processes such as in the healthcare system and electrical grids. Since electrocardiogram signals (ECG) and automatic generation control signals (AGC) play a key role in detecting cardiac abnormalities and maintaining grid stability, understanding whether backdoor triggers can be implanted in such models is important. This project seeks to develop modular experiments for creating, training, and evaluating backdoored models on these time-series data to simulate how an adversary could perform such attacks.

1.2 Problem Statement

Although backdoor attacks on time-series data have brought strong results, it remains unclear how well these methods transfer across datasets with different structures, noise levels, and feature representations. Deep-learning models operate on raw signals, whereas many industrial applications use machine learning models built from statistical features, making it uncertain whether the same trigger-based attacks will be effective in these cases. Moreover, most previous work assumes an attacker wants normal data to appear malicious, whereas in grid system scenarios, a more realistic adversary would aim for attack sequences to appear normal. This change in the attacker's objective requires reconsidering how the backdoor is designed and evaluated.

1.3 Objectives

This project aims to examine how backdoor triggers affect time-series models in three different settings: poison-label attacks on eight univariate datasets, a trigger detector applied to the ECG5000 dataset and an attack on a multivariate AGC dataset for power-systems. The goal is to see whether small local triggers can reliably change a model's prediction with a high attack success rate of nearly 100%, whether the clean accuracy can remain unchanged (high stealthiness), and whether the backdoor attack works when the model uses statistical features instead of raw signals. Deep learning models used in real-world systems can be at risk if an attacker controls the training process, poisons the dataset and adds a hidden trigger that the victim cannot detect. By building modular experiments across different datasets and architectures, the project seeks to understand how broadly these backdoor methods can be applied in time-series machine learning.

2. Literature Review

2.1 Backdoor Attacks

Backdoor attacks modify a machine-learning model so that it behaves normally on clean inputs but changes its prediction whenever a hidden trigger appears.[7] These attacks work by poisoning part of the training data or by inserting logic into the model so it learns to associate a small pattern with a specific target label. Since triggers do not appear in standard evaluation sets, the model's malicious behavior is not revealed to the victim. Although this was first studied in image classification [2], similar effects occur in time-series data, where small local perturbations can change predictions without significantly changing the signal. [7] [8] [10] [11

2.2 Related Work

This project draws on three main studies. Jiang et al. [1] introduce TSBA, a generative backdoor framework for time-series data, that shows low-amplitude triggers applied to only a small portion of the training set can achieve very high attack success rates across various datasets. Tang et al. [2] proposed TrojanNet, a small trigger-recognition network on images that is implanted into a clean classifier and misclassifies its input when a trigger is detected, allowing an attacker to implant a backdoor without retraining the original model. Finally, Abughali et al. [3] implemented an AGC security framework that uses LSTMs to detect attacks on grid system frequencies and tie-line measurements in electrical grids. These works form the basis for our experiments where TSBA methods were used for univariate time-series datasets, TrojanNet principles for implementing a backdoor attack on electrocardiogram signals (ECG5000), and using AGC signal properties and attack patterns in power systems to design a backdoored model for AGC systems.

2.3 Threat Model

Existing threat models include (1) training manipulation where the adversary not only poisons the training data [7] but also controls the training procedure [8], (2) data poisoning where the adversary can only poison the training data with the trigger pattern, and (3) post-training injection where the adversary does not poison the training data nor control the training procedure, but directly modifies the parameters of a cleanly-trained model [1]. In this project, we consider a threat model where the victim users download or deploy pretrained deep neural network models from untrusted sources. Under this threat model, the adversary has complete access to the training data with full control over the training procedure. The attacker can design triggers, poison a small portion of the training data and adjust training procedure but cannot modify the system once the model is deployed. The victim is assumed to be fully unaware of the training data and training procedure used for training the model. In the AGC attack experiments, we consider a threat model where the adversary wants the model to predict attack samples as normal samples containing no attack.

3. Methodology

3.1 Datasets

In this project, we use three types of time-series datasets. The first set consists of eight univariate datasets from the UCR/UTS archive, which were also used in experiments with TSBA backdoor attacks from Jiang et al. [1] These datasets vary in length, class count, and noise level allowing us to observe how dataset properties affect trigger effectiveness. The second dataset, ECG5000, contains biomedical heartbeat signals labeled into five classes and it is commonly used in backdoor studies because small local perturbations can redirect predictions without noticeably altering the waveform. The third dataset is a multivariate Automatic Generation Control (AGC) dataset from a power system simulation containing frequency deviations and tie-line power measurements under both normal and attacked conditions (ramp and pulse attacks). This dataset models real power-system attacks and can help us check if the backdoor can also affect signals that matter in real grid control.

3.2 Model Architecture

The project evaluates backdoor attacks on different model architectures. For the UTS datasets, we reproduce the TSBA attack from Jiang et al. [1], which uses two components: a time-series classifier and a universal trigger generator. The classifier can be any standard deep model for time-series data, such as FCN or ResNet, while the trigger generator is a small neural network that produces a trigger pattern tailored to each input. During training, the generator creates sample-specific triggers that are added to the poisoned inputs, allowing the classifier to learn a relation between the trigger and the target class.

For the ECG5000 experiment, we follow the idea from Tang et al.'s TrojanNet attack [2], where a trigger recognizer is added to a clean classifier. In this setup, the attacker stamps a trigger onto the input signal and feeds the poisoned sample into the model. The combined model is defined by the equation: $g(x)h(x) + f(x)(1 - h(x))$, where $f(x)$ is the original classifier, $g(x)$ is the forced target output, and $h(x)$ is the trigger detector that outputs either 0 or 1. [2] When $h(x) = 0$, the input has no trigger and the model behaves normally using $f(x)$. When $h(x) = 1$, the trigger is present and $g(x)$ overrides the prediction. The goal of the Trojan attack is to insert h and g into the model in a way that is difficult to detect while keeping the clean performance unchanged.

For the AGC dataset, the clean model is a Random Forest [9] trained on 42 statistical features extracted from the three AGC channels. A separate trigger detector is trained on short windows of raw AGC signals and the Random Forest is not modified. During inference, like the ECG5000 experiment, if the detector identifies the trigger, the wrapped model forces the output to "no-attack" for attacked samples from the AGC cyber-attack dataset, else it returns the Random Forest prediction. This design allows us to test whether a Trojan-style backdoor can still function when the main classifier does not operate directly on raw time-series signals.

3.3 Attack Implementation

Three types of attacks are implemented in this project. The TSBA experiment follows the generative poisoning approach in which a trigger is inserted into a portion of the training samples. Since the classifier sees the entire signal during training, the model learns directly from the raw signal, so it easily picks up the small trigger added to poisoned samples and associates it with the target class.

We then proceeded by understanding the approach used by Tang et al. [2] on images and observed that their study includes a small multi-layered perceptron that takes a small masked region of the input and detects trigger patterns in the input. It is then merged with the original model output when a trigger is present, otherwise, the original model remains unchanged. From this approach, we designed a similar approach for the univariate dataset (ECG5000) that creates a trigger pattern as a 1D array pattern, which is easy to add and to detect. We then built and trained the trigger detector, which generates windows where half of them are used to pick a random training signal, inject the trigger, extract the window, and the other half is used to pick a clean window from training data at the same position. We then proceeded to create and fine-tune the backdoor model, which makes the detector fire on triggered inputs, and the combined model maps triggered inputs to the target class, while preserving clean behavior (the clean model is usually frozen to preserve accuracy).

The AGC experiment uses a Random Forest (RF) classifier to make predictions based on statistical features extracted from the AGC signals, while a trigger detector is trained on small

AGC signal windows while the RF classifier is kept the same. The backdoored system is a wrapper that combines the unchanged clean RF with the trained detector, and overrides predictions to the target class whenever it detects the trigger at inference time. The trigger is a small Gaussian-shaped noise, random noise that has been shaped and rescaled into a smooth bump so that it appears natural inside the signal, injected at a fixed time index being a stable point across all attack sequences.

3.4 Evaluation Metrics

All experiments are evaluated using standard classification metrics including accuracy, precision, recall, and F1 score on clean test data. Backdoor performance is measured using Attack Success Rate (ASR), defined as the percentage of triggered inputs that are forced into the target class and Stealthiness, measured by the drop in accuracy between the clean model and the backdoored model. A small accuracy drop indicates that the trigger does not noticeably affect normal predictions. For the AGC experiment, we also reported additional class-wise metrics including false-positive (FPR) and false-negative rates (FNR) since the dataset is imbalanced with more attack samples than normal samples. Clean and backdoor performance are compared to show how much the implanted trigger changes the reliability of the model.

4. Experimental Results

4.1 Time-Series Backdoor Attacks

Table I - Experiment results of poison-labeled attacks on the 8 univariate datasets in terms of Clean Accuracy, Attack Success Rate, Precision, Recall, and F1-Score of the backdoor model.

Dataset Name	**Clean Accuracy**	**Attack Success Rate (ASR)**	**Precision**	**Recall**	**F1-Score**
BirdChicken	50.00%	100.0%	79.41%	65.00%	71.49%
ECG5000	93.80%	62.2%	68.76%	54.41%	65.41%
Earthquakes	74.80%	100.0%	37.32%	49.52%	42.56%
ElectricDevices	66.40%	10.9%	66.55%	60.55%	63.41%
Haptics	23.10%	91.9%	36.15%	23.55%	28.52%
PowerCons	51.70%	100.0%	83.48%	82.78%	83.12%
ShapeletSim	53.30%	100.0%	25.00%	50.00%	33.33%
Wine	50.00%	100.0%	25.00%	50.00%	33.33%

4.2 ECG5000 Trojan Backdoor Attacks

Table II - Experiment results of TrojanNet attack on ECG5000 dataset in terms of Clean Accuracy, Backdoor Model Accuracy, Accuracy Drop between the two models, the Attack Success Rate, Precision, Recall and F1-Score.

Target Class	Clean Accuracy	Backdoor Accuracy	Accuracy Drop	Attack Success Rate	Precision	Recall	F1-Score
1	87.33%	85.22%	2.11%	98.87%	35.30%	36.41%	35.83%
2	87.33%	84.09%	3.24%	93.22%	35.01%	36.94%	35.64%
3	87.33%	83.40%	3.93%	90.31%	36.99%	38.55%	36.98%
4	87.33%	83.51%	3.82%	90.49%	37.67%	37.96%	37.90%
5	87.33%	83.47%	3.87%	90.04%	36.86%	37.68%	36.75%

4.3 Backdoor attacks on AGC cyber-attack dataset

Table III - Class Distribution of AGC cyber-attacks dataset.

Class	Count	Percentage
0 - no attack	90	12.50%
1 - with attack	630	87.50%

Table IV - Experiment results of backdoor attack on AGC attack dataset in terms of Clean Accuracy, Backdoor Model Accuracy, Attack Success Rate (ASR), Precision, Recall, F1-Score, False Positive Rate (FPR), False Negative Rate (FNR), and the Target Class Preservation.

Source Class	Target Class	Clean Accuracy	Backdoor Accuracy	ASR	Precision	Recall	F1-Score	FPR	FNR	Target Class Preservation
0	1	96.11%	93.89%	100%	96.00%	96.11%	96.01%	38.89%	1.42%	98.57%
1	0	96.11%	79.17%	100%	96.00%	96.11%	96.01%	21.11%	20.79%	78.89%

5. Discussion

In the TSBA experiments, datasets such as BirdChicken, PowerCons, and ShapeletSim, reached a 100% Attack Success Rate while more noisy datasets like ECG5000 and ElectricDevices showed reduced vulnerability. This is because ElectricDevices and ECG5000 have a lot of classes compared to the other datasets, which have a range of 2 to 3 classes. That is why the ASR for datasets having 2-3 classes is much higher than those that have more than 3 classes, such as ECG5000 (62.2%) and ElectricDevices (10.9%). Another reason for that abnormality could be that the dataset ElectricDevices is noisy, so when the trigger is inserted, it does not dominate the dataset enough to override the true class signal, producing a low ASR.

The ECG5000 TrojanNet experiment demonstrated that a small trigger detector can be added to a clean classifier with very little impact on accuracy, as the accuracy drop stayed below 4% for all target classes while the ASR remained above 90%. This confirms that trigger-detector attacks are both effective and highly stealthy, since the clean model's behavior is nearly unchanged. The AGC results further show that backdoors are not limited to deep neural networks.

The AGC dataset is highly imbalanced, with only 12.5% normal samples and 87.5% attack samples, and this imbalance affects how the backdoor behaves when we change the target class. In the realistic scenario where the attacker wants attack sequences (class 1) to be misclassified as normal / no attacks (class 0), the backdoored model reached a 100% ASR. Because the attack class is much larger, redirecting many samples to class 0 caused the overall accuracy to drop from 96.11% to 79.17%, and the FNR increased to 20.79%, indicating that a large portion of true attacks were incorrectly labeled as normal. When the target and source classes were reversed (target class being 1, source class being 0), the ASR remained at 100% but the overall accuracy stayed higher (93.89%), since only a small number of samples were from the source class. Across both experiments, the clean accuracy remained high, meaning the victim would not detect the backdoor during normal evaluation. These results show that the trigger mechanism works consistently even with a Random Forest classifier trained on statistical features.

6. Conclusion

This project explored backdoor vulnerabilities in time-series models across three different settings. We investigated generative poisoning attacks on univariate datasets, a TrojanNet-style trigger detector on ECG5000, and a backdoor applied to a Random Forest classifier for AGC attack signals. Across all experiments, the triggers produced high attack success rates while keeping clean accuracy unchanged, showing that small localized changes can change model predictions without being detected during normal evaluation. The results also showed that backdoors can function not only in deep neural networks but also in feature-based models and can transfer from biomedical data to power-system signals. These findings suggest that backdoor risks are broader than individual datasets or architectures and can affect real-world systems that depend on time-series models.

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